LOSTLINK

Secure Digital Lost-and-Found System

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Project: LostLink – Secure Digital Lost-and-Found System
Department: Computer Science and Engineering (CSE)

This page contains the project team information provided by the user and is included as part of the complete project report.

LOSTLINK

Secure Digital Lost-and-Found System

Project Concept	A centralized platform for reporting, discovering, matching, and securely recovering lost and found belongings.
Primary Goal	Reduce the time and effort required to reunite people with their lost items while preventing false or unauthorized claims.
Core Security Idea	A simple verification question is used before an item can be claimed.

Prepared from the supplied project brief
This document expands the four requested sections: Problem Statement, Proposed Solution/Approach, Key Features, and Wireframe/Solution Flow.

Contents.

Reference brief captured exactly

The supplied brief defines LostLink as a secure digital lost-and-found system. Its core features are: posting lost or found items, searching by category or description, matching similar lost and found posts, and using a simple verification question before claiming.

1. Problem Statement

LostLink is proposed as a secure digital lost-and-found system that addresses the difficulty people face when they lose or discover belongings in places such as colleges, offices, apartments, public facilities, events, or shared communities.

The supplied problem brief states: **“Build a secure digital lost-and-found system.”** The brief specifically requires four core capabilities: **post lost or found items; search items by category or description; match similar lost and found posts; and use a simple verification question before claiming.**

In a conventional lost-and-found process, information may be scattered across notice boards, messaging groups, social media posts, reception desks, or verbal communication. This makes it difficult to know what has already been reported, whether an item has been found, and whether the person requesting it is the genuine owner.

The problem is therefore not only **finding an item**; it is also **organizing information, connecting related reports, and making the recovery process trustworthy**.

1.1 Major Problems

- **Scattered reporting:** Lost and found information can exist in multiple places, making it difficult to discover relevant reports.
- **Slow discovery:** A person may repeatedly ask others whether an item was found instead of searching one centralized system.
- **Incomplete descriptions:** Users may describe the same object differently, making manual matching difficult.
- **Duplicate effort:** Multiple people can create separate posts about the same incident or item.
- **Unauthorized claims:** Without verification, someone could falsely claim an item that belongs to another person.
- **No clear status:** Users need to know whether an item is still lost, found, under verification, returned, or closed.
- **Privacy concerns:** Sensitive contact details and identifying information should not be unnecessarily exposed to everyone.

1.2 Problem Objective

The objective is to create one secure, searchable platform where a user can report a lost or found item, discover relevant posts, receive possible matches, and complete a lightweight ownership verification process before an item is handed over.

1.3 Target Users

Person who lost an item	Creates a lost-item report, searches found reports, reviews possible matches, and answers the verification question to prove ownership.
Person who found an item	Creates a found-item report, reviews possible lost-item matches, communicates through the system, and verifies the claimant before returning the item.

1.4 Example Scenario

Example: Lost College ID Card

A student loses an ID card on campus. Instead of posting in several messaging groups, the student creates a LostLink post with category = ID Card, approximate location = Library, date = today, and a description. Later, another student finds an ID card and creates a Found post. LostLink identifies the two posts as potentially related. Before the card is returned, the claimant must answer a verification question about a detail that was not publicly revealed in the post.

2. Proposed Solution / Approach

LostLink solves the problem through a centralized workflow that combines structured reporting, search and filtering, similarity-based matching, secure communication, and claim verification.

The system should treat every lost/found report as a structured record rather than just free-form text. This makes reports easier to search, compare, filter, and manage.

2.1 Overall Approach

1	**Create an account / sign in:** The user accesses the system through a basic authenticated account.
2	**Post a report:** The user selects Lost or Found and enters item details such as title, category, description, date, approximate location, and optional image.
3	**Store and index the report:** The system stores the report with a unique ID, user reference, timestamps, and status.
4	**Search and filter:** Users can search by category, keywords, description, location, date, and Lost/Found type.
5	**Generate possible matches:** The system compares relevant attributes between lost and found reports and presents likely matches.
6	**Review a candidate:** The user opens the candidate post and checks visible details without exposing hidden verification information.
7	**Start a claim:** The claimant requests the item and must complete the verification step.
8	**Verify ownership:** The finder/authorized administrator asks a simple question based on a hidden item detail. A correct answer increases confidence that the claimant is genuine.
9	**Complete return:** Once verified, the parties arrange the handover through the approved process and the report is marked Returned/Resolved.

2.2 Logical Architecture

Presentation Layer	Application Layer	Data Layer
Web/mobile interface • Login • Dashboard • Post forms • Search • Match results • Claim screen	Authentication Report management Search/filter logic Matching engine Claim workflow Notifications Authorization	Users Reports Categories Matches Claims Verification questions Messages Audit/status history

2.3 Matching Approach

A practical first version can use a weighted rule-based score. The score compares category, keywords in the description, location, date/time proximity, and optional image similarity. This is transparent and easy to explain during a project demonstration.

Illustrative matching score: Category match 30% + description/keyword similarity 30% + location similarity 20% + date/time proximity 15% + optional image similarity 5%. The weights can be tuned after testing.

For example, a lost “black wallet” reported near the cafeteria today should rank a found “black leather wallet” reported at the cafeteria today above an unrelated found phone reported elsewhere. The system should display matches as **possible matches**, not as guaranteed ownership.

2.4 Secure Claim Approach

The verification question is intentionally simple but should use information that is **not fully visible in the public post**. For example, a finder may record a hidden detail such as the exact sticker inside a wallet, the last four digits printed on a card, or a unique mark. The claimant must provide the correct answer before the claim proceeds.

Important security principle

Do not publish the answer to the verification question in the public item description. The question should help distinguish the genuine owner from someone who is simply copying public information.

3. Key Features

3.1 User Authentication

Users can register and sign in so that each report and claim is associated with a known account. Authentication reduces anonymous abuse and supports accountability.

3.2 Post Lost Item

A user selects “Lost” and provides item name/title, category, description, approximate location, date/time, optional image, and additional identifying information. The user can save a hidden verification detail for later ownership verification.

3.3 Post Found Item

A finder selects “Found” and records the item details, where it was found, when it was found, and optional image. The finder should avoid publishing every unique identifying detail.

3.4 Category-Based Search

Users can filter by categories such as Electronics, Documents, Wallets, Keys, Bags, Clothing, Accessories, Books, IDs, or Other. Categories make browsing faster and more consistent.

3.5 Description / Keyword Search

A search bar allows users to search item names and descriptions. Keyword normalization can help treat terms such as “mobile,” “phone,” and “smartphone” as related where appropriate.

3.6 Filters

Useful filters include Lost/Found, category, date range, approximate location, status, and sorting by newest or most relevant.

3.7 Similar Post Matching

The system compares lost and found posts and generates possible matches. A match card can show why it was suggested, for example: same category, similar description, nearby location, and close date.

3.8 Verification Before Claiming

Before an item is handed over, the claimant must answer a verification question based on a hidden detail. The result can be recorded as Verified, Failed, or Pending.

3.9 Claim Management

A user can submit a claim against a found item. The finder or authorized administrator can review the request, ask the verification question, approve/reject the claim, and update the item status.

3.10 Status Tracking

Every report can have a clear state such as Lost, Found, Match Suggested, Claim Pending, Verified, Returned, or Closed.

3.11 Notifications

Optional notifications can inform users when a possible match appears, someone submits a claim, a claim is approved, or an item status changes.

3.12 Privacy-Preserving Contact

Instead of exposing personal phone numbers or email addresses publicly, the system can provide controlled communication or reveal contact details only after an appropriate stage.

3.13 Reporting / Moderation

Users can report suspicious, duplicate, offensive, or fraudulent posts. An administrator can review flagged content and take action.

3.14 Admin Dashboard

An administrator can view reports, users, claims, flagged content, unresolved cases, and audit history. Admin controls should be permission-based.

3.15 Core Feature Summary

Feature	Purpose	Security / Quality Benefit
Lost/Found Posting	Creates structured reports	Consistent, traceable records
Search & Filters	Find relevant posts quickly	Reduces time and duplicate effort
Matching	Connects related lost/found posts	Improves recovery chances
Verification	Checks claimant knowledge	Reduces false claims
Status Tracking	Shows current case state	Prevents outdated actions
Admin Controls	Handles moderation and disputes	Improves accountability

4. Wireframe / Solution Flow

The following wireframes are conceptual layouts. They describe what each major screen should contain and how a user moves through the system. They can later be converted into a high-fidelity UI or directly implemented in a web/mobile application.

4.1 Home / Dashboard Wireframe

LOSTLINK	Search ■	Login / Profile
Find a lost item Search found posts and possible matches.	**Report an item** Lost | Found	**My Activity** My posts | Claims | Status
Filters Category • Location • Date • Status	**Recent Posts** Cards with image, title, location, date, status	**Possible Matches** Relevant suggestions

4.2 Post Item Wireframe

Report Item	Select type: ■ Lost ■ Found
Item title	e.g., Black Wallet
Category	Electronics / Documents / Wallet / Keys / Other
Description	Color, brand, visible features, approximate contents, unique marks
Location	Where it was lost/found (approximate location)
Date & time	When it was lost/found
Image	Optional item photo
Private verification detail	Hidden detail used to verify the genuine claimant
	[Submit Report]

4.3 Search & Match Wireframe

Search Lost & Found	■ Search: black wallet
Filters	**Results**

Type: Lost / Found Category: Wallet Location: Cafeteria Date: Today Status: Active	**Possible Match #1** Black leather wallet • Cafeteria • Today Match reasons: same category + similar description + nearby location **Possible Match #2** Black wallet • Library • Yesterday

4.4 Claim & Verification Wireframe

Claim Item	Verification	Result
Selected found item Claimant submits request	Finder/authorized user asks: “What is the hidden identifying detail?”	Correct → Verified → Return process Incorrect → Claim remains pending/rejected

4.5 End-to-End Solution Flow

User	Lost/Found Report	System	Finder/Owner	Resolution
Register / Login	—	Authenticate user	—	Account ready
Submit Lost/Found post	Create structured report	Store + index	—	Report becomes searchable
Search / Browse	Select filters	Return relevant posts	—	Candidate items displayed
Review match	Open candidate	Calculate/display match reasons	—	Possible match identified
Claim item	Submit claim	Create claim record	Review request	Verification starts
Answer question	Provide hidden detail	Record result	Ask verification question	Verified / failed
Handover	Arrange return	Update status	Return item	Returned / Closed

5. System Roles and User Journey

5.1 User Roles

Role	Responsibilities
Registered User	Create and manage lost/found posts, search items, view matches, submit claims, and track status.
Finder	Post found items, review claim requests, conduct verification, and complete the return process.
Owner / Claimant	Search for the lost item, identify a possible match, submit a claim, and answer the verification question.
Administrator	Moderate users/posts, handle disputes, manage flagged content, review suspicious activity, and maintain system records.

5.2 Lost-Item User Journey

1	User logs in and selects **Report Lost Item**.
2	User enters item details and saves a private verification detail.
3	System validates required fields and creates the report.
4	The report becomes searchable and the matching engine looks for relevant found reports.
5	User receives or sees possible matches.
6	User opens a promising found report and selects **Claim**.
7	The system creates a claim request and starts the verification process.
8	Finder/admin asks the verification question; claimant answers.
9	If verified, the finder and claimant proceed with the approved handover process.
10	The report status changes to **Returned/Closed**.

5.3 Finder Journey

1	Finder logs in and selects **Report Found Item**.
2	Finder records the item, location, date/time, description, and optional photo.
3	Finder stores a hidden identifying detail that is not published publicly.
4	System makes the found report searchable and looks for possible lost-item matches.

5	When a claim arrives, finder reviews the claimant and asks the verification question.
6	If the answer is correct, finder marks the claim verified and completes the return process.
7	Finder updates the report so other users know that the item is no longer available.

5.4 Administrator Journey

1	Review dashboard for active reports, claims, flags, and unresolved cases.
2	Check suspicious or duplicate reports.
3	Handle users or content that violate system rules.
4	Support disputes where a normal verification flow is insufficient.
5	Review audit history and ensure resolved items are correctly marked.

6. Data and Security Design

6.1 Suggested Data Entities

Entity	Important Fields	Purpose
User	user_id, name, email, password_hash, role, created_at	Identifies authenticated users and permissions.
Item Report	report_id, user_id, type, title, category, description, location, event_time, image, status	Stores each lost/found report.
Match	match_id, lost_report_id, found_report_id, score, reasons, created_at	Stores possible relationships between reports.
Claim	claim_id, report_id, claimant_id, status, created_at, verified_at	Tracks a request to claim an item.
Verification	verification_id, claim_id, question, answer_hash/secure value, attempts, result	Supports ownership verification.
Message	message_id, sender_id, receiver_id, claim_id, content, timestamp	Provides controlled communication.
Audit Log	log_id, user_id, action, entity, timestamp, metadata	Supports accountability and investigation.

6.2 Security Requirements

- **Password protection:** Never store plain-text passwords. Store strong password hashes using a modern password-hashing method.
- **Authentication:** Require a valid session/token for protected actions such as creating claims or changing report ownership/status.
- **Authorization:** A normal user should not be able to edit another user's report or access administrator functions.
- **Private verification information:** Hidden answers should never be exposed in public APIs, search results, or ordinary item cards.
- **Rate limiting:** Limit repeated claim attempts and verification attempts to reduce abuse.
- **Input validation:** Validate text, image type/size, dates, and required fields to prevent malformed or malicious input.
- **Safe file handling:** Uploaded images should be validated and stored securely rather than trusting the filename or content type.
- **Privacy:** Show only the information needed at each stage. Avoid publishing personal contact details unnecessarily.
- **Auditability:** Record important events such as post creation, claim creation, verification outcome, status changes, and administrator actions.
- **Secure transport:** Use HTTPS in production so credentials and claim-related communication are protected in transit.

6.3 Verification Rules

Good verification question	Weak verification question
Asks about a detail specific to the genuine owner and not completely disclosed in the public post. Example: “What color is the small sticker inside the wallet?”	Asks something obvious from the public listing. Example: “What color is the wallet?” when the post already says “black wallet.”

The verification process should be treated as a supporting security measure, not as absolute proof of ownership. For high-value or sensitive items, an administrator or additional evidence may be required.

7. Functional Requirements and Use Cases

7.1 Functional Requirements

FR-01	The system shall allow users to register and authenticate.
FR-02	The system shall allow a user to create a Lost report.
FR-03	The system shall allow a user to create a Found report.
FR-04	The system shall capture category, description, location, date/time, and status for each report.
FR-05	The system shall allow users to search by category and description/keywords.
FR-06	The system shall support filtering by report type and other relevant attributes.
FR-07	The system shall generate possible matches between related Lost and Found reports.
FR-08	The system shall allow a claimant to submit a claim against a Found report.
FR-09	The system shall require verification before a claim is approved.
FR-10	The system shall maintain item and claim status.
FR-11	The system shall provide role-based administrative controls.
FR-12	The system should preserve an audit trail for important actions.

7.2 Non-Functional Requirements

- **Usability:** Posting and searching should be simple enough for a first-time user.
- **Performance:** Search results should load quickly for normal usage.
- **Reliability:** Reports and claim records should not be lost when the system is restarted.
- **Security:** Protected data and operations must be accessible only to authorized users.
- **Scalability:** The design should allow the number of reports and users to grow without redesigning the entire application.
- **Maintainability:** Matching, authentication, reporting, and claim logic should be separated into manageable modules.

7.3 Main Use Cases

Use Case	Actor	Outcome
Register / Login	User	Authenticated session created.
Post Lost Item	Owner	Lost report becomes searchable.
Post Found Item	Finder	Found report becomes searchable.
Search Items	User	Relevant reports are displayed.
View Possible Match	User	Potential lost/found relationship is shown.

Claim Item	Claimant	Claim request is created and awaits verification.
Verify Claim	Finder/Admin	Claim is verified or rejected.
Return Item	Finder + Claimant	Item is handed over and report is closed.
Moderate Content	Admin	Suspicious/invalid content is reviewed.

8. Expected Benefits, Limitations and Future Scope

8.1 Expected Benefits

- **Faster recovery:** A centralized search and matching process can reduce the time spent manually checking many channels.
- **Higher visibility:** Lost and found reports are available in one searchable location.
- **Better organization:** Categories, statuses, timestamps, and locations make reports easier to manage.
- **Reduced false claims:** Verification adds an extra barrier against people claiming items they cannot genuinely identify.
- **Less manual coordination:** Matching and notifications can reduce repetitive communication.
- **Traceability:** Status and audit records provide a history of what happened to a report or claim.

8.2 Limitations

- A verification question cannot guarantee legal ownership; it is a lightweight practical check.
- Matching quality depends on the accuracy and completeness of user-provided descriptions.
- Image-based matching may produce false positives or miss visually similar items.
- Users may fail to update the status after an item is returned.
- High-value or sensitive items may require stronger identity and administrative verification.

8.3 Future Enhancements

- **AI-powered semantic matching:** Use embeddings or an LLM-based similarity layer to understand descriptions beyond exact keywords.
- **Image similarity:** Compare uploaded item images to improve candidate ranking.
- **Location maps:** Display approximate lost/found locations without exposing overly precise private information.
- **QR-based handover:** Generate a one-time handover code that both parties confirm during return.
- **Smart notifications:** Alert users only when a high-confidence match appears.
- **Duplicate detection:** Detect repeated reports describing the same item.
- **Community moderation:** Add trusted users or institutional moderators to improve safety.
- **Institution integration:** Connect LostLink with college, office, hostel, apartment, or event administration systems.
- **Analytics:** Provide dashboards showing recovery rate, average resolution time, popular categories, and unresolved cases.

8.4 Suggested Development Phases

Phase	Scope
Phase 1 – MVP	Authentication, Lost/Found posting, category/description search, basic status management.
Phase 2 – Matching	Possible-match engine using category, keywords, location, and date proximity.

Phase 3 – Secure Claims	Claim requests, verification questions, controlled communication, audit logs.
Phase 4 – Intelligence	Semantic matching, image similarity, smart notifications, duplicate detection.
Phase 5 – Scale	Administration, analytics, institutional integration, performance and security hardening.

9. Conclusion

Final solution summary

LostLink centralizes lost-and-found reporting, makes items easy to search, automatically highlights possible matches, and adds a verification step before claiming. This creates a faster, more organized, and more secure recovery experience for owners, finders, and administrators.

Recommended project demonstration: Demonstrate two accounts (owner and finder), create a lost post and a found post, show the match, submit a claim, answer the hidden verification question, and finally change the item status to Returned/Closed.